

COMP 3311: Database Management Systems

Lab 2 Exercise: Oracle Database, SQL*Plus and Oracle SQL Developer

WHAT TO DO

1. **Download** the zipped folder **Lab2ExerciseFiles.zip** from the *Lab 2* entry of the *Lab Assignments* section on Canvas to the desktop and unzip it. The folder contains the script file **lab2db.sql** which contains SQL statements that:
 - drop (delete) a table named **Student** if it exists;
 - create a table named **Student** with six attributes;
 - inserts 22 different records into the **Student** table.Don't worry if you do not understand the SQL statements. They will be covered in detail in future labs and the lectures.

2. **Execute** the **lab2db.sql** script file in **Oracle SQL Developer**.

Note: The following **SQL*Plus** command is used in the **lab2db.sql** script file.

clear screen clears the Script Output pane.

3. **Create** a new SQL script file in **Oracle SQL Developer** named **insertMyself.sql**.

4. **Construct**, in the **insertMyself.sql** script file, an SQL **insert** statement that inserts into the **Student** table a record with the following values. (See the **lab2db.sql** script file for examples of such an insert statement.)
 - Your student id (8 digits max), first name (20 characters max), last name (25 characters max) and HKUST email login name **excluding** "@connect.ust.hk" (15 characters max).
 - Any phone number of your choosing; it **does not** have to be your phone number (8 digits max).
 - Any valid year for the **admissionYear** attribute (4 digits).
 - The value "3.56" for the **cga** attribute.
 - The value "COMP" for the **unitId** attribute.

Note: All values, except for **cga**, are strings and need to be enclosed in single quotes.

Add the command **commit** as the second (and last) line of the script file. This command writes any changes you make to a table from main memory to disk (i.e., to the database).

5. **Save** your **insertMyself.sql** script file inside the **Lab2Exercise** folder.
6. **Execute** your **insertMyself.sql** script file in **Oracle SQL Developer** using the **Run Script** button.
7. **Open** an SQL worksheet and in this worksheet **construct** and **execute** an SQL **select** statement to show all the records in the **Student** table as shown in Figure 1.

	STUDENTID	FIRSTNAME	LASTNAME	EMAIL	PHONENO	ADMITYEAR	CGA	UNITID
1	13455789	Harry	Potter	cspotter	23581234	2023	2.76	COMP
2	15456789	Leo	Da Vinci	csdavinci	23585678	2023	2.72	COMP
3	13556789	Leo	Greenleaf	magreenleaf	23582468	2024	3.36	MATH
4	13456789	Ariana	Grande	csgrande	23581234	2024	2.82	COMP
5	15678989	Edith	Clarke	csclarke	23589876	2024	2.73	COMP
6	15678901	Albert	Einstein	cseinstein	23585678	2023	2.97	COMP
7	16789012	Robert	Redford	maredford	23582468	2024	2.57	MATH
8	14567890	David	Decker	eedecker	23589876	2024	1.49	ELEC
9	99987654	Lazy	Lazy	cslazy	23581357	2024	(null)	COMP
10	26184624	Bruce	Wayne	eewayne	28261057	2023	2.47	ELEC
11	26184444	Donald	Trump	bstump	28255057	2024	1.49	MGMT
12	26186666	Warren	Buffet	bsbuffet	28266027	2023	3.42	MGMT
13	28435018	David	Decker	bsdecker	27619435	2024	1.49	MGMT
14	66666666	Ferris	Bueller	bsbueeller	28282727	2023	1.54	MGMT
15	15000655	Steve	Jobs	csjobs	26232244	2023	3.56	COMP
16	15085942	Bill	Gates	csgates	25678679	2024	3.4	COMP
17	28834512	Isaac	Newton	manewton	22861987	2023	2.98	MATH
18	28918856	Alan	Turing	maturing	26679834	2023	3.56	MATH
19	29873381	Nikola	Tesla	eetesla	25671983	2023	3.37	ELEC
20	13782973	Edith	Clarke	eeclarke	28340180	2023	3.15	ELEC
21	18792018	Elon	Musk	bsmusk	28659910	2024	2.76	MGMT
22	27182481	Buzzy	Bizy	bsbizy	26178891	2024	(null)	MGMT
23	11111111	Typical	Student	typical	91919191	2023	3.56	COMP

Your information should be shown in place of the highlighted record.

Figure 1: Example **Oracle SQL Developer** Query Result tab showing **Student** records.

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WHAT TO SUBMIT

1. Your **insertMyself.sql** script file containing only the following two statements:
 - a. the SQL **insert** statement that adds your student record and
 - b. the **commit** command.
2. A jpg/jpeg screenshot file or PDF file of the **Oracle SQL Developer** Query Result tab showing the result of the SQL **select** statement as shown in Figure 1.

HOW TO SUBMIT

By 6:00 p.m. tomorrow, upload your **insertMyself.sql** script file and jpg/jpeg screenshot file or PDF file to Canvas by selecting *Lab 2* in the *Lab Assignments* section of Canvas, and then selecting the **Submit Assignment** button. You need to upload each file separately. To check your submission, select the **Submission Details** button on the right side of Canvas. For help, select the **Help** button at the top-right of Canvas.

WHAT TO SAVE

Save your **insertMyself.sql** script file as it will be needed in subsequent labs.